

Welcome to NRES 746

Advanced Analysis Methods in Natural Resources and Environmental Science

Fall 2025

Course Information

Instructor Info

Kevin Shoemaker

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Office hours: Mondays after class at 1pm or by appointment

Course Meeting Times

Lecture: M, W at noon in FA 109 (50 mins)

Lab: Tuesday at 3pm in TBA (2 hours 45 mins)

Course Website

<https://kevintshoemaker.github.io/NRES-746/>

Course Objectives

Our general focus will be on *model-based inference*, including linear and non-linear regression, multi-level models, and multi-model inference. But instead of using existing software tools, we build our own **custom (bespoke) models** and make inferences using maximum likelihood and Bayesian methods. You should emerge from this course as a creative data analyst with the tools and intuition needed to draw your own conclusions using complex and messy data.

In this course, we are mathematicians. Many of us do not consider ourselves very math-savvy, and some of us shut down when we see a mathematical equation. But it's impossible to overstate the value of mathematics for logical reasoning, problem solving and unambiguously communicating correct solutions to others. There are no mathematics prerequisites for this course, but we will frequently use concepts from multivariable calculus and linear algebra, since this is the real language of probability and statistics. Therefore, we will learn to write down our models in the simplest and most unambiguous form: mathematical equations.

In this course, we are computer programmers. A focus on custom (bespoke) methods for inference allows us to focus on our study systems and research questions, instead of being constrained by the high-level software tools available to us. Mathematical equations give us the language of describing our custom models, but the solutions quickly become intractable to closed-form mathematical solutions. This is where the computer takes over- computational methods can be designed to get us pretty darn close to the target solutions- and that is usually more than enough. We just have to know a little programming to take advantage of these incredibly powerful methods.

This is a safe space to learn and be dangerous Sometimes our research questions or intricacies of our study system lead us to dream up models that our 'canned' software can't handle easily. So it is very powerful to be able to *build custom algorithms*. When we do this, we are entering

uncharted territory. And exploring can be dangerous... and your inner voices might tell you not to go there, just play it safe. Don't listen to those voices! In this class, you are allowed - and encouraged - to *be dangerous*! Just remember not to apply this motto uncritically outside the classroom!! It can be exciting to apply novel models to data! Remember: science is require integrity and rigor above all. Therefore, we need to move slowly and cautiously- especially when they are trying something totally new!!

(NOTE: where appropriate, I encourage you to use well-tested, standard software tools because these methods have been rigorously tested over the years.)

Student Learning Outcomes

Upon completion of this course, students will be able to:

1. Identify and contrast the major classes of statistical methods used by natural resources scientists (e.g., Bayesian vs frequentist, likelihood-based, non-parametric methods, machine learning) and explain some of the key advantages and disadvantages of these methods.
2. Apply analysis tools such as logistic regression, non-linear regression, hierarchical (multi-level) models, and multivariate analyses on diverse data sets representative of those commonly encountered in the natural resources sciences.
3. Learn to explore data sets quantitatively and graphically and to prepare data for analysis.
4. Perform command-line programming operations, statistical analysis, data visualization, and simulation modeling with the statistical computing language 'R'.
5. Critically evaluate the strength of inferences drawn from statistical models by testing assumptions and assessing performance.
6. Communicate concepts and methods via student-led lectures and discussion on advanced topics in data analysis.

Prerequisites

Curious scientific mind, broad research interests, and a willingness to engage with equations and computer code. Students are expected to already have a basic foundation in mathematical and statistical concepts and methods, obtained through other coursework. If this is not the case, they should be prepared to work harder to develop the necessary prerequisite knowledge.

Textbooks and Readings

We will use the book, *Ecological Models and Data in R*, by Ben Bolker, as a general class reference.

Another recommended text (which I will draw some examples from and may serve as a primary text in the future) is *Statistical Rethinking* by Richard McElreath.

Additional readings may be assigned -please check the course schedule.

Other related books you may find useful include "A Primer of Ecological Statistics" by Gotelli and Ellison, "Bayesian Models: a Statistical Primer for Ecologists" by Hobbs and Hooten, and the textbooks on Bayesian hierarchical models by Kery and Royle.

Course components

Class Participation: Students are expected to actively participate in the classroom. Don't be afraid to ask questions- this is difficult material, and this classroom is a safe space for exploring and making mistakes!

Lab Reports: For each lab unit, students will submit (1) an R script ('.R' file, submitted via WebCampus) that performs a specific assigned task, and (2) short answer responses (also submitted via WebCampus). While students are encouraged to work on the labs in small groups, all lab submissions are made individually.

Group Projects: Students will work on projects in groups of 2 - 3. Projects will require analysis of previously published or publicly available data sets that are NOT intended to be part of a student's thesis or dissertation chapters. The instructor can assist with identifying suitable data sets. Although a primary goal is to enhance knowledge and facility with the data analysis methods, an important secondary goal could be to develop a collaborative manuscript for publication! Therefore, careful thought should go into choice of a data set and relevant scientific questions to guide the analysis. The group project will take the form of a standard research manuscript.

Group projects: expectations

Students are expected to perform (and write up the results for) a data analysis using a custom, model-based analysis. The write-up will loosely take the format of a scientific paper to be submitted to a professional journal. However, because of the nature of this course, the most important pieces of the write-up are the methods and results sections. Nonetheless, I expect at least a few paragraphs introducing the topic and why it's important, and a few paragraphs discussing the implications of the results. The methods and results section can (and in many cases should) be much longer than you typically see in a scientific paper- don't feel constrained by space for these sections! Not that you need to be wordy, I just want to make sure you have the space to clearly explain the analyses you performed and why you made the choices you did!

Here is a more detailed description of expectations for the final group project:

Introduction: Provide enough description so that the reader (me!) understands why the research is important and (if appropriate) what hypotheses or ideas are being tested.

Methods: Provide some details about data collection, just enough to give the reader the context necessary to understand the data. Provide plenty of detail about the analytical approach- enough detail to fully replicate the analysis. Justify all decisions that were made and (where appropriate) discuss why you did not use alternative approaches. Discuss key analytical assumptions. Use equations!! This section can be longer than the methods section of a standard manuscript.

Results: Present all relevant results completely and concisely. There is a limit of 5 figures and 3 tables, so choose carefully which figures and tables to present. Figures should be publication quality.

Discussion: Write at least three paragraphs that put the results in a larger context and discusses areas of uncertainty. Potential topics are possible violations of assumptions, and future work that your analysis suggests would be profitable.

Code and data Provide all code (and ideally the raw data) used to run the analyses presented in the paper in a GitHub repository.

Grading

Course component	Weight
Laboratory exercises	40%
Participation	15%
Research Project, proposal	5%
Research Project, written	30%
Research Project, presentation	10%

Letter grade assignment is as follows: A (93 to 100%), A- (90 to 92.9%), B+ (87 to 89.9%), B (83 to 86.9%), B- (80 to 82.9%), C+ (77 to 79.9%), C (73 to 76.9%), C- (70 to 72.9%).

Course Schedule

NOTE: the course schedule is subject to change, so please check back frequently at

<https://kevintshoemaker.github.io/NRES-746/schedule.html>

Week	Lecture 1	Lab	Lecture 2	Material Covered	Readings
Aug. 25	Course Introduction	Lab #1: Custom algorithms in R	Custom algorithms	Review syllabus, programming algorithms in R	Clark Ch. 1; Touchon and McCoy 2016; Bolker Ch 1
Sept. 1	No class (labor day)	Lab #1: Custom algorithms in R	Custom algorithms	Basic concepts in statistics, probability	Bolker ch. 4
Sept. 8	Probability	Lab #2: "Virtual Ecologist"	Probability	Probability, mechanistic models, stochastic models	Bolker Chs. 2, 3
Sept. 15	Data Generating Models	Lab #2: "Virtual Ecologist"	Data Generating Models	Power analysis, goodness-of-fit testing, likelihood	Bolker Ch. 5; Hobbs and Hilborn 2006 (optional)
Sept. 22	Likelihood	Lab #3: Maximum likelihood	Likelihood	Likelihood theory, maximum likelihood	Bolker Ch. 6
Sept. 29	Likelihood	Lab #3: Maximum likelihood	Optimization	Maximum likelihood inference, optimization algorithms	Bolker Chs. 6, 7

Oct. 6	Optimization	Group meetings: final projects	Bayesian inference	General introduction to Bayesian theory and application	Bolker Ch. 7 (Bayesian section); Ellison 2004
Oct. 13	Bayesian inference	Lab #4: Bayesian model fitting	Markov Chain Monte Carlo (MCMC)	Markov-Chain Monte Carlo for Bayesian inference	Bolker Ch. 7 (Bayesian section);
Oct. 20	Model selection and feature elimination	Lab #4: Bayesian model fitting	Model selection re: Bayesian inference	Model selection using information theoretic criteria	Bolker Ch. 6 (AIC section)
Oct. 27	Model performance evaluation	Student project progress reports and discussion	Model performance evaluation	Bias-variance tradeoff, cross-validation, assessing predictive accuracy	
Nov. 3	Multilevel models	Demo/activity: multilevel models	Multilevel models	Multilevel models, "Random effects", non-independence	
Nov. 10	Bayesian multilevel models	Group meetings: final projects	Bayesian multilevel models	Bayesian multilevel (hierarchical) models	
Nov. 17	Nonlinear regression, and GAMs	Demo/activity: GAMs in 'mgcv'	Nonlinear models and GAMs	Generalized additive models (GAMs)	
Nov. 24	Spatial models	Demo/activity: Spatial models in INLA	Time-series models	Spatial autocorrelation	
Dec. 1	Time-series models	Demo/activity: time-series model in 'stan'	Multivariate statistical models	Temporal autocorrelation	
Dec. 8	Class wrap-up- where to go from here	Student project presentations!	No class (prep day)		
Dec. 15	NA (classes over)				

Class Procedures

Students are expected to attend all lectures and labs in person. The instructor will attempt to record lectures and post to WebCampus for review- but attendance and engagement in class/labs is critical for learning. All assignments and labs should be submitted using WebCampus. Deadlines for assignments can be found on WebCampus.

Make-up policy and late work:

If you miss a class meeting or lab period, it is your responsibility to talk to one of your classmates about what you missed. If you miss a lab meeting, you are still responsible for completing the lab activities and write-up by the stated due date. Please let your instructor know in advance if you need to miss class or lab.

University Policies

Statement on Academic Dishonesty

The University Academic Standards Policy defines academic dishonesty, and mandates specific sanctions for violations. See the University Academic Standards policy: UAM 6,502

Statement of Disability Services

Any student with a disability needing academic adjustments or accommodations is requested to speak with the instructor or the Disability Resource Center (Pennington Achievement Center Suite 230) as soon as possible to arrange for appropriate accommodations. This course may leverage 3rd party web/multimedia content, if you experience any issues accessing this content, please notify your instructor.

Statement on Audio and Video Recording

Surreptitious or covert video-taping of class or unauthorized audio recording of class is prohibited by law and by Board of Regents policy. This class may be videotaped or audio recorded only with the written permission of the instructor. In order to accommodate students with disabilities, some students may have been given permission to record class lectures and discussions. Therefore, students should understand that their comments during class may be recorded.

Class sessions may be audio-visually recorded for students in the class to review and for enrolled students who are unable to attend live to view. Students who participate with their camera on or who use a profile image are consenting to have their video or image recorded. If you do not consent to have your profile or video image recorded, keep your camera off and do not use a profile image. Students who un-mute during class and participate orally are consenting to have their voices recorded. If you do not consent to have your voice recorded during class, keep your mute button activated and only communicate by using the “chat” feature, which allows you to type questions and comments live.

This is a safe space

The University of Nevada, Reno is committed to providing a safe learning and work environment for all. If you believe you have experienced discrimination, sexual harassment, sexual assault, domestic/dating violence, or stalking, whether on or off campus, or need information related to immigration concerns, please contact the University’s Equal Opportunity & Title IX office at 775-784-1547. Resources and interim measures are available to assist you. For more information, please visit the Equal Opportunity and Title IX page.

Statement on Campus Closures or Delays

In the event of class cancellations or delays caused by inclement weather conditions, fire/smoke conditions, or other unforeseen emergencies, the safety and well-being of students

are the University's top priority. Official notifications will be disseminated through the University website and other official channels with details related to any campus delays or closures. In the event of a campus closure, you will be informed as to whether the class will be offered remotely or if it will be canceled. If the class is cancelled, you will receive information on how to address any missed course content. Students facing significant impacts due to these events are encouraged to communicate with their instructor for potential accommodations.

Statement on Failure to Comply with Policy (including as outlined in this Syllabus) or Directives of a University Employee:

In accordance with section 6,502 of the University Administrative Manual, a student may receive academic and disciplinary sanctions for failure to comply with policy, including this syllabus, for failure to comply with the directions of a University Official, for disruptive behavior in the classroom, or any other prohibited action. "Disruptive behavior" is defined in part as behavior, including but not limited to failure to follow course, laboratory or safety rules, or endangering the health of others. A student may be dropped from class at any time for misconduct or disruptive behavior in the classroom upon recommendation of the instructor and with approval of the college dean. A student may also receive disciplinary sanctions through the Office of Student Conduct for misconduct or disruptive behavior, including endangering the health of others, in the classroom. The student shall not receive a refund for course fees or tuition.